How (and why) to work as an open scientist in the 21st century

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Slides: http://bit.ly/eglen-todos (CC-BY license)

Declarations

1. Affiliate editor of *bioRxiv*
2. Senior editor of *Scientific Data*

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About me

My research and how I got into open science.

What is open science

- My definition of open science: a way to approach research which promotes sharing of (all / most) research artifacts.
- From this, open access (OA) to research papers is normally assumed. But we all make mistakes...

My #OAfail

Sharing your data is easier than you think

Stephen Eglen

Nature **510**, 340(2014) | Cite this article

457 Accesses | **3** Citations | **43** Altmetric | Metrics

Geoffrey Goodhill questions some of the practicalities of open data-sharing policies (*Nature* **509**, 33; 2014), but I believe that his concerns are largely unfounded.

Storing large volumes of raw data is costly, but many items destined for sharing are highly processed and relatively small. The mouse-brain connectome, for example, is available as a 3-megabyte file derived from many gigabytes of raw data (S. W. Oh *et al. Nature* **508**, 207–214; 2014). Neither is there a shortage of repositories: many institutional databases are freely available and well supported (such as zenodo.org, maintained by CERN, Europe's particle-physics lab in Geneva, Switzerland). More repositories will come online as researchers learn how to share data more effectively.

Contrary to Goodhill's suggestion, sharing computer code does not necessarily demand much time investment (see, for example, D. C. Ince *et al. Nature* **482**, 485–488; 2012). Code is a valuable part of a paper, so everyone benefits if its authors assume from the start that it will be shared or reused. Also, people releasing code are under no obligation to maintain it.

r im("eglen2014_metrics1.png", 200)`

When to submit a preprint?

Estimates from https://nikokriegeskorte.org/2016/03/13/the-selfish-scientists-guide-to-preprint-posting/

Moral or selfish approach to sharing?

Markowetz *Genome Biology* (2015) 16:274
DOI 10.1186/s13059-015-0850-7

Genome Biology

COMMENT Open Access

Five selfish reasons to work reproducibly

Florian Markowetz

Abstract

And so, my fellow scientists: ask not what you can do for reproducibility; ask what reproducibility can do for you! Here, I present five reasons why working reproducibly pays off in the long run and is in the self-interest of every ambitious, career-oriented scientist.

Keywords: Reproducibility, Scientific career

how science actually is. And, whether you like it or not, science is all about more publications, more impact factor, more money and more career. More, more, more… so how does working reproducibly help me achieve more as a scientist.

Reproducibility: what's in it for me?

In this article, I present five reasons why working reproducibly pays off in the long run and is in the self-interest of every ambitious, career-oriented scientist.

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Selfish reasons to share

Why not align what is good for science with what is good for scientists?

1. Funding mandates (REF + enforcement from Wellcome Trust)
2. Credit through data papers
3. Fixes data bugs / errors in analysis
4. Prevent data loss (Vines et al 2014). e.g. students have a habit of leaving...
5. Your future self is probably one of the main beneficiaries of sharing.
6. *Now* is a very good time to be an open scientist.
7. Leads to further collaborations
8. Reviewers can do more work...

Rule 1: Data should be shared

Given the cost of generating data, I think data relating to a publication should be shared along with the paper.

Funders (and increasingly many, but not all, journals) agree.

Pregistration studies

Evaluate journal subscription model

Think about which journals you support, by sending papers and reviewing.

Support OA journals as authors and reviewers. (Ever asked a journal for permission to republish one of your own figures?)

Check out status of journal via Sherpa/Romeo service.

What can your institutions & funders do?

- sign up to DORA (Declaration of Research Assessment) (Wellcome Trust have solved this problem). Signing is easy, enforcing is hard.
- Give ECR a voice within institutions.
- Recognise all research artefacts, not just papers, as valuable outputs.
- Give academics more ownership of funds. "Do you really want to spend 5K on an APC, versus other uses in your lab?"
- Routes to OA need to be generate less admin for librarians / funders, not more. Green OA achieves that.
- Support diamond OA initiatives led by academics and academic societies.

Conclusions

- CODECHECK sets the bar very low to encourage adoption
- Looking for feedback on how to get journals to adopt this.